

Statistics Hybrid Practice Set — S17

Focus

S17 moves from basic interpretation toward **real Data Analyst decision-making**. No Q1 population/sample question and no formula-listing question.

Q1 — Mean vs Median Business Decision

A company records monthly customer spending:

₹800, ₹850, ₹900, ₹950, ₹1,000, ₹1,100, ₹6,500

Answer:

1. Calculate the mean.
 2. Calculate the median.
 3. Which measure should the company use to describe **typical customer spending**?
 4. Explain why the other measure could be misleading.
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Q2 — Detect the Better Measure

Two datasets describe employee salaries:

Dataset	Mean	Median
A	₹52,000	₹51,000
B	₹75,000	₹48,000

Answer:

1. Which dataset shows stronger evidence of extreme high salaries?
2. Which measure would you use to represent typical salary for Dataset B?
3. Why?

Q3 — Spread Comparison

Two delivery teams have:

Team	Mean Delivery Time	SD
A	40 min	4 min
B	40 min	12 min

Answer:

1. Which team has greater variability?
 2. Which team is more consistent?
 3. What does the difference in SD tell you about the delivery times?
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Q4 — CV Decision

Three stores have:

Store	Mean Sales	SD
A	₹50,000	₹4,000
B	₹80,000	₹8,000
C	₹1,20,000	₹9,600

Calculate the CV for each store.

Then:

1. Which store has the lowest relative variability?
 2. Rank the stores from **most consistent** → **least consistent**.
 3. Are the stores equally consistent? Explain.
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Q5 — Quartile Interpretation

A dataset has:

Statistic	Value
Minimum	12
Q1	20
Median	28
Q3	40
Maximum	75

Answer:

1. Calculate the IQR.
 2. What percentage of observations lie between 20 and 40?
 3. Which half of the middle 50% has greater spread?
 4. Is the median closer to Q1 or Q3?
 5. What does this suggest about the distribution?
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Q6 — Outlier Detection

Dataset:

10, 12, 14, 15, 18, 20, 21, 23, 25, 60

Using the IQR method:

1. Find Q1.
2. Find Q3.
3. Find IQR.
4. Find the lower fence.
5. Find the upper fence.
6. Is 60 an outlier?
7. Explain your conclusion.

Q7 — Box Plot Comparison

Two departments have these five-number summaries:

Statistic	Dept A	Dept B
Minimum	15	10
Q1	25	20
Median	30	35
Q3	40	55
Maximum	50	80

Answer:

1. Which department has the higher median?
 2. Calculate the IQR for both.
 3. Calculate the range for both.
 4. Which department has greater variability?
 5. Which department appears more consistent?
 6. Which department has a longer upper tail?
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Q8 — Identify the Skewness

Three datasets have the following relationships:

Dataset A

Mean = 50

Median = 50

Dataset B

Mean = 65

Median = 50

Dataset C

Mean = 40

Median = 50

Identify:

1. Which dataset is approximately symmetric?
 2. Which dataset is likely right-skewed?
 3. Which dataset is likely left-skewed?
 4. Explain the relationship between mean and median that helped you decide.
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Q9 — Data Analyst Method Selection

Choose the **best statistical measure or visualization** for each situation.

A.

A company wants to identify unusually high and unusually low employee salaries.

B.

A company wants to compare consistency between two products whose average prices are very different.

C.

A manager wants to report the typical house price when a few luxury houses exist.

D.

An analyst wants to compare the distributions of five different departments on one visual.

E.

An analyst wants to measure the spread of the middle 50% of customer spending.

Choose from:

- Mean
 - Median
 - IQR
 - CV
 - Box plot
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Q10 — Full Data Analyst Case

A food-delivery company compares two delivery partners:

Partner	Mean	Median	SD
A	32 min	28 min	8 min
B	45 min	44 min	5 min

Answer:

1. Calculate CV for both partners.
2. Which partner has lower **absolute variability**?
3. Which partner has lower **relative variability**?
4. Which partner appears more right-skewed?
5. If the company wants **more consistent delivery relative to average delivery time**, which partner should it choose?
6. Give **one reason why looking only at SD would lead to an incomplete decision**.